

Install the OpenSSH server:

```
sudo yum install openssh-server
```

Step 3: Start and Enable the SSH Service

Start the SSH service:

```
sudo systemctl start sshd
```

Enable the service to start on boot:

```
sudo systemctl enable sshd
```

Check the status of the SSH service:

```
sudo systemctl status sshd
```

Step 4: Configure the Firewall (if applicable)

If a firewall is active, allow SSH connections:

```
sudo firewall-cmd --permanent --add-service=ssh
```

```
sudo firewall-cmd --reload
```

Step 5: Generate SSH Key Pair

Generate an SSH key pair for secure authentication:

```
ssh-keygen:
```

- This is the command-line utility used to generate, manage, and convert SSH keys. It creates a public-private key pair for use with SSH authentication.

- `-t rsa`:

- The -t option specifies the type of key to create.

In this case, rsa stands for Rivest-Shamir-Adleman, which is a widely used public key encryption algorithm. RSA is known for its security and is commonly used for SSH keys.

- -b 4096:

- The -b option specifies the number of bits in the key.

4096 means the key will be 4096 bits long, which provides a high level of security.

A longer key length makes it more difficult for attackers to break the encryption through brute-force methods.

- -C "your_email@example.com":

- The -C option allows you to add a comment or label to the key, typically used to identify the key's purpose or owner.

In this case, "your_email@example.com" is a placeholder for your actual email address. This comment can help you recognize the key later, especially if you generate multiple keys.

```
ssh-keygen -t rsa -b 4096 -C "your_email@example.com"
```

Press Enter to accept the default location (~/.ssh/id_rsa), and optionally enter a passphrase for added security.

Step 6: Copy the Public Key to the Authorized Keys

Create the .ssh directory and the authorized_keys file (if it doesn't exist):

```
mkdir -p ~/.ssh touch ~/.ssh/authorized_keys
```

Copy your public key to the authorized_keys file:

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

Set the correct permissions for the .ssh directory and authorized_keys file:

```
chmod 700 ~/.ssh
```

```
chmod 600 ~/.ssh/authorized_keys
```

Step 7: Test the SSH Connection Locally

Open a new terminal window (or tab) in your VM.

Connect to the server (itself) using SSH:

```
ssh username@localhost
```

Replace username with your actual username. You should be able to log in without being prompted for a password if you set up key-based authentication correctly.